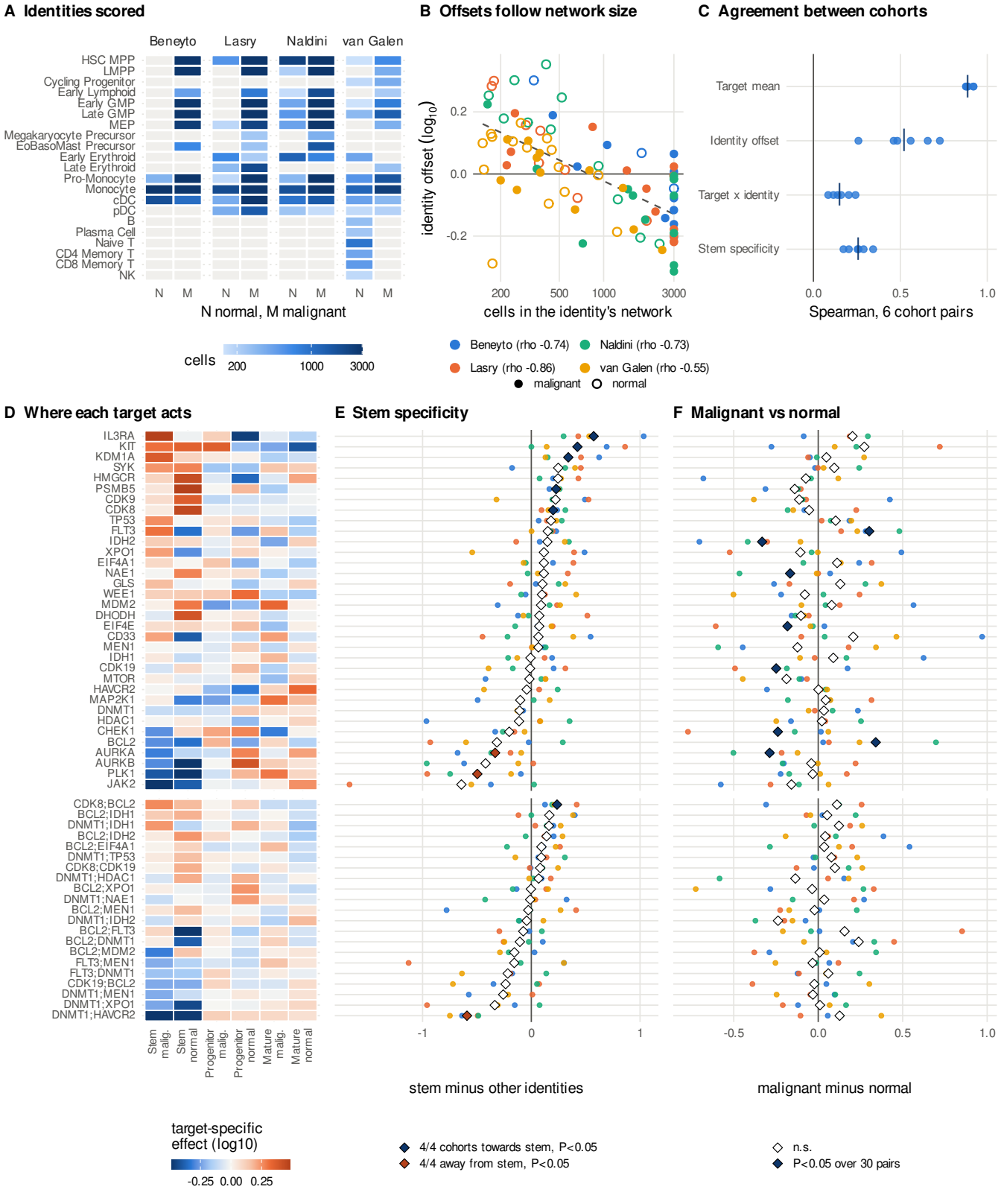

A Identities scored
Beneyto
Lasry
Naldini
van Galen
HSC MPP
LMPP
Cycling Progenitor
Early Lymphoid
Early GMP
Late GMP
MEP
Megakaryocyte Precursor
EoBasoMast Precursor
Early Erythroid
Late Erythroid
Pro-Monocyte
Monocyte
cDC
pDC
B
Plasma Cell
Naive T
CD4 Memory T
CD8 Memory T
NK
N M N M N M N M
N normal, M malignant
cells
200
1000
3000
B Offsets follow network size
identity offset (log10)
0.2
0.0
-0.2
200
500
1000
3000
cells in the identity's network
C Agreement between cohorts
Target mean
Identity offset
Target x identity
Stem specificity
0.0
0.5
1.0
Spearman, 6 cohort pairs
Beneyto (rho -0.74)
Naldini (rho -0.73)
Lasry (rho -0.86)
van Galen (rho -0.55)
malignant
normal
D Where each target acts
IL3RA
KIT
KDM1A
SYK
HMGCR
PSMB5
CDK9
CDK8
TP53
FLT3
IDH2
XPO1
EIF4A1
NAE1
GLS
WEE1
MDM2
DHODH
EIF4E
CD33
MEN1
IDH1
CDK19
MTOR
HAVCR2
MAP2K1
DNMT1
HDAC1
CHEK1
BCL2
AURKA
AURKB
PLK1
JAK2
CDK8;BCL2
BCL2;IDH1
DNMT1;IDH1
BCL2;IDH2
BCL2;EIF4A1
DNMT1;TP53
CDK8;CDK19
DNMT1;HDAC1
BCL2;XPO1
DNMT1;NAE1
BCL2;MEN1
DNMT1;IDH2
BCL2;FLT3
BCL2;DNMT1
BCL2;MDM2
FLT3;MEN1
FLT3;DNMT1
CDK19;BCL2
DNMT1;MEN1
DNMT1;XPO1
DNMT1;HAVCR2
Stem malig.
Stem normal
Progenitor malig.
Progenitor normal
Mature malig.
Mature normal
target-specific effect (log10)
-0.25 0.00 0.25
E Stem specificity
-1
0
1
stem minus other identities
4/4 cohorts towards stem, P<0.05
4/4 away from stem, P<0.05
F Malignant vs normal
-0.5
0.0
0.5
1.0
malignant minus normal
n.s.
P<0.05 over 30 pairs